Using Disease Surveillance and Response to Facilitate Adaptation to Climate-Related Health Risks

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Surveillance

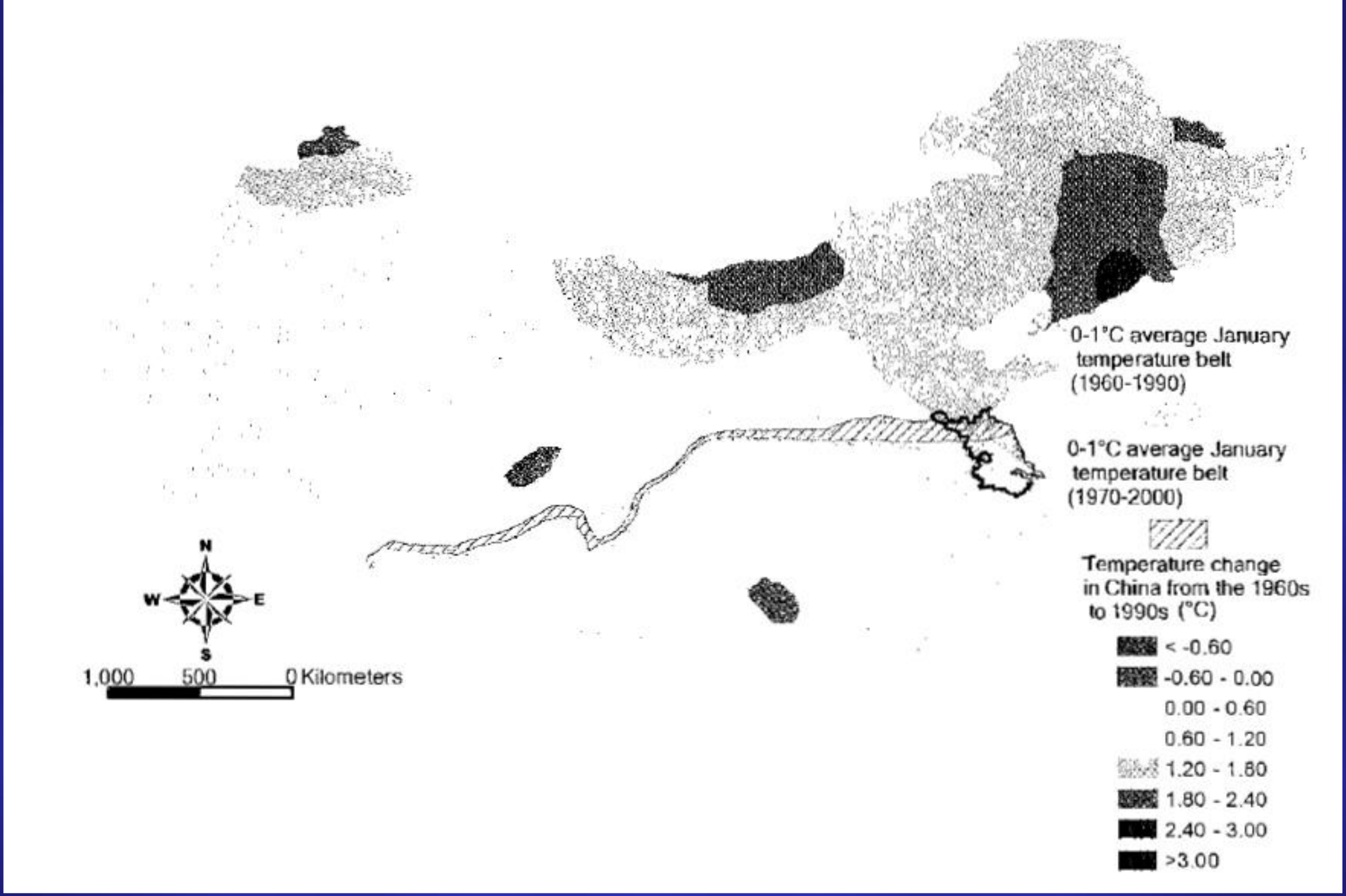
- The continuing scrutiny of all aspects of the occurrence and spread of health outcomes that are pertinent to effective control
  - Guided by the need to make public health decisions in the context of constant vigilance and pragmatic responses
- Involves systematic collection of health, disease, and environmental information, and the interpretation and distribution of this information to relevant actors
  - Includes risk factors necessary to interpret disease data

Some Features of Surveillance

- Effective public health infrastructure required
  - Financial and human capital costs of surveillance play a key role in determining the type and quantity of data collected
    - Laboratories required to verify diagnoses, conduct quality assurance, etc.
- Accuracy and timeliness important
- Spatial and temporal scales of health and environmental data need to be matched
- Legal and ethical framework important
- Context and practice of surveillance vary greatly from one country to another

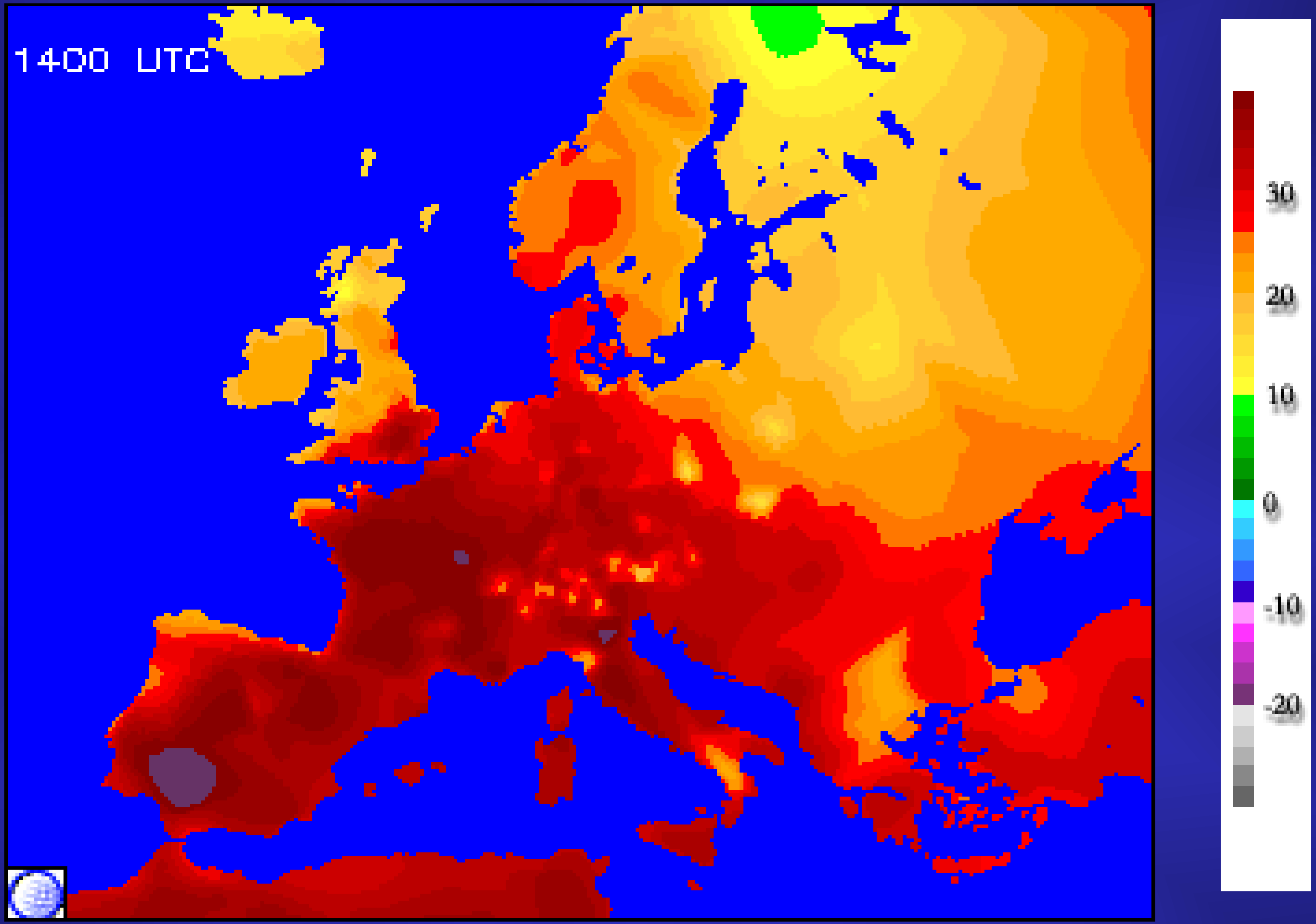
Wilson and Anker 2005

Average January Temperature Difference Between the 1960s and 1990s, and Additional Area at Risk for Schistosomiasis



Yang et al. 2003

Maximum Temperature
August 10, 2003

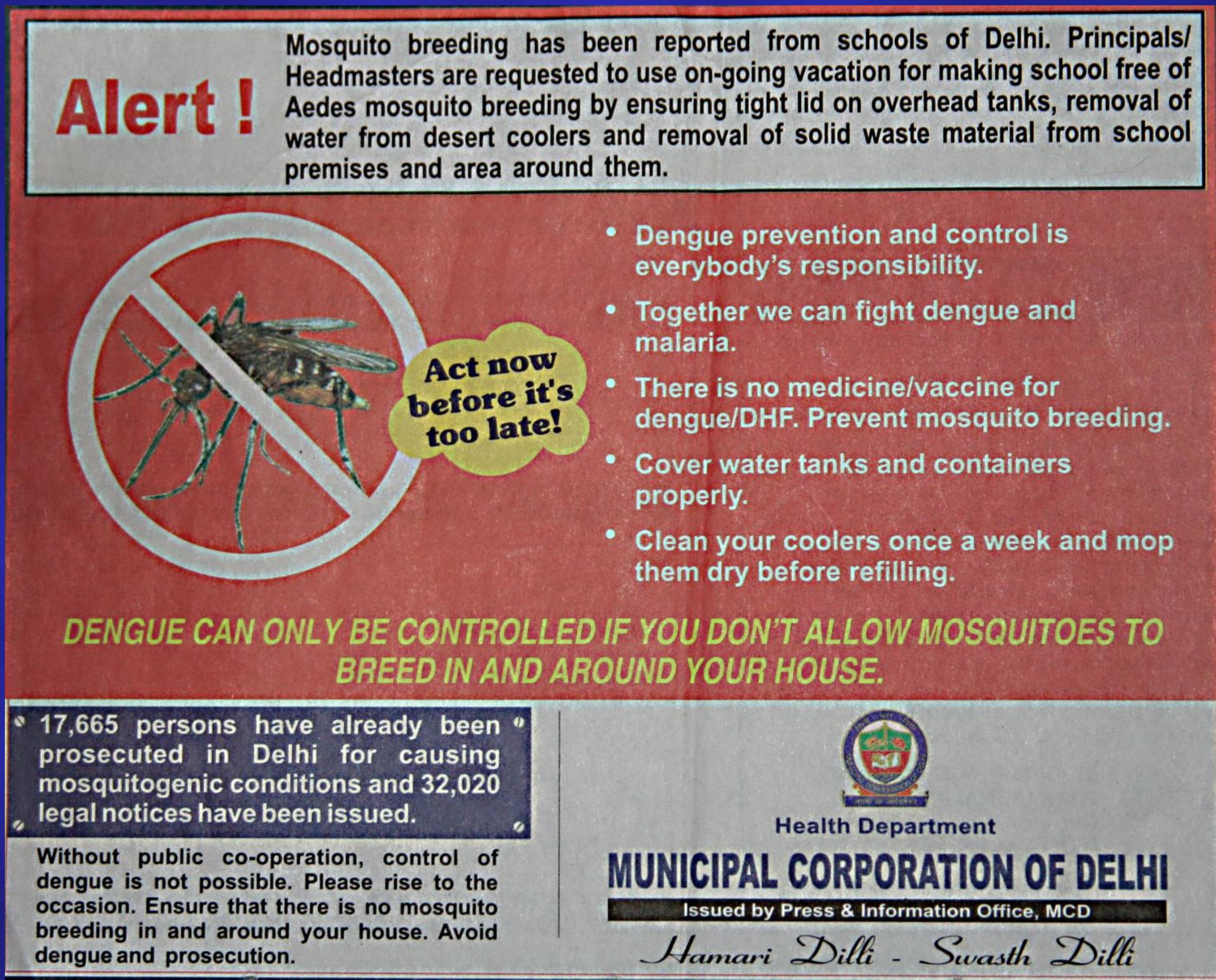


Components of a Response Plan

- Where the response plan will be implemented
  - Need to consider influence of climate change
- When interventions will be implemented, including thresholds for action
- What interventions will be implemented
- How the response plan will be implemented
- To whom the interventions will be communicated

Ebi and Schmier 2005

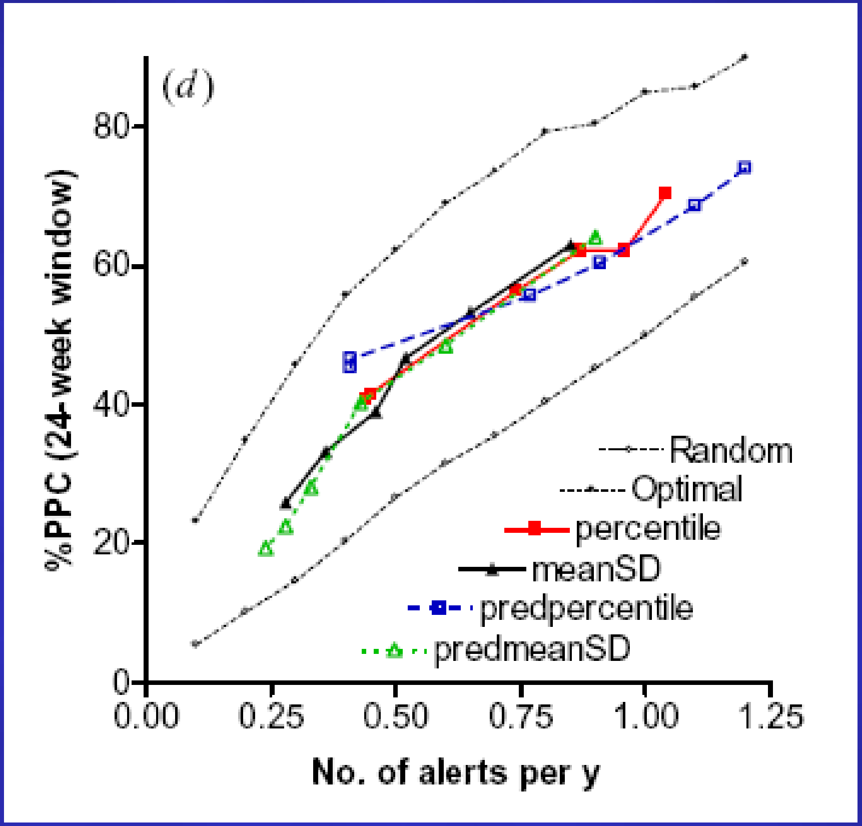
Effective Interventions are Embedded in an Understanding of Human Factors and Address Local Situations



Construction of Predictive Models of Health Outcomes

- Evaluation of potential for epidemic transmission
- Identification of epidemic-prone areas and populations at risk to allow rapid:
  - Prediction and detection
  - Targeting of response
  - Planning of logistics for response
- Quantification of climatic and non-climatic disease risk factors
- Quantification of the links between climate, other factors, and disease outbreaks to construct model

Weather-Based Prediction of *Plasmodium falciparum* Malaria in Ethiopia: Comparison with Early Detection



Teklehaimanot et al. 2004

Monitoring and Evaluation

- Need to establish programs to answer these questions (at a minimum):
  - What are the chances that we will fail to predict an epidemic, and how many lives would be lost?
  - What are the chances of sounding a false alarm, thereby wasting resources and undermining public trust?
  - Is the system as responsive as needed? How many lives could have been saved if the system response was faster?
  - Is the system cost-effective?

Candidate Diseases for Epidemic Early Warning Systems

- Cholera
- Malaria
- Dengue
- Japanese encephalitis
- Influenza
- Others

Thank You